

JEE MAIN REVISION MASTER: BINOMIAL THEOREM SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 160 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Binomial Theorem - Core Application	160 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Binomial Theorem - Core Application (Very High Yield — 160 PYQs)

Core Principle: Key problem pattern in Binomial Theorem.

■ ■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "If $26(\frac{2^3}{3}({}^{12}C_2) + \frac{2^5}{5}({}^{12}C_4) + \frac{2^7}{7}({}^{12}C_6) + \dots + \frac{2^{13}}{13}({}^{12}C_{12})) = 3^{13} - a \dots$ " [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Let the smallest value of $k \in \mathbb{N}$, for which the coefficient of x^3 in $(1+x)^3 + (1+x)^4 + (1+x)^5 + \dots + (1+x)^{99} + (1+kx)^{100}, x \dots$ " [Ans: B]
- [2026 Morning Shift] Q#1 (MCQ): "Let $S = (1/25!) + (1/3!23!) + (1/5!21!) + \dots$ up to 13 terms. If $13S = \frac{2^k n!}{k!}, k \in \mathbb{N}$, then $n+k$ is equal to..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 4th April Morning Shift (Q#1)

Let the smallest value of $k \in \mathbb{N}$, for which the coefficient of x^3 in $(1+x)^3 + (1+x)^4 + (1+x)^5 + \dots + (1+x)^{99} + (1+kx)^{100}$, $x \neq 0$, is $(43n + (101/4)) \binom{100}{3}$ for some $n \in \mathbb{N}$, be p . Then the value of $p+n$ is :

- (A) 10
- (B) 11 **[Correct]**
- (C) 12
- (D) 13

Correct Answer: B

Solution Analysis: Coefficient of x^3 in the given expression $(1+x)^3 + (1+x)^4 + \dots + (1+x)^{99} + (1+kx)^{100}$ is obtained by adding the coefficients of x^3 from each term. For $(1+x)^r$, coefficient of x^3 is $\binom{r}{3}$. So, from $(1+x)^3 + (1+x)^4 + \dots + (1+x)^{99}$ the total coefficient of x^3 is $\sum_{r=3}^{99} \binom{r}{3}$. Also, in $(1+kx)^{100}$, coefficient of x^3 is $\binom{100}{3} k^3$. Hence total coefficient of x^3 is $\sum_{r=3}^{99} \binom{r}{3} + \binom{100}{3} k^3$. Now use the standard NCERT identity: $\sum_{r=3}^{99} \binom{r}{3} = \binom{100}{4}$. So coefficient becomes $\binom{100}{4} k^3$.